Human Activity Recognition using Smartphone & Smartwatch Sensors

COMP4220 – Machine Learning

University of Massachusetts Lowell

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Introduction & Problem Statement

What is Human Activity Recognition?

Definition

Automatic classification of human activities from wearable sensor data (accelerometer, gyroscope)

Why Classify Activities?

Enable context-aware services, personalized experiences, health monitoring, and safety applications

Sensors Used

Accelerometer and gyroscope data from smartphone and smartwatch devices

Real-World Applications

- Fitness Tracking
- Fall Detection
- Rehabilitation
- Smart Homes
- Activity Coaching
- Health Monitoring

Motivation & Objective

Motivation

Wearables Growth

Smartwatches and smartphones are ubiquitous, worn by millions of users daily

Continuous Data Stream

Devices produce continuous multi-sensor data (accelerometer, gyroscope) capturing user movements

Opportunity

Rich sensor data enables intelligent activity recognition and context-aware applications

Project Objective

Primary Goal

Classify user activity from smartphone and smartwatch sensor data with high accuracy

Approach

Use 94 engineered features extracted from accelerometer and gyroscope time windows

Model Comparison

Compare multiple machine learning models and select the best performer

Expected Outcome

Identify optimal ML approach for real-world wearable activity recognition systems

Dataset Overview: WISDM

DATASET

WISDM

Smartphone & Smartwatch Activity Recognition

TOTAL SAMPLES

75,099

Activity instances

FEATURES

94

Accelerometer + Gyroscope

DEVICES

2 Types

Smartphone + Smartwatch

ACTIVITY CLASSES

Top 10

Balanced selection

Preprocessing Pipeline

1. **Load and Merge ARFF Files**
   Import smartphone and smartwatch sensor data from ARFF format

2. **Clean Column Names**
   Standardize feature names and remove inconsistencies

3. **Select Top 10 Balanced Classes**
   Filter and balance activity classes for optimal training

4. **StandardScaler Normalization**
   Scale features to zero mean and unit variance

5. **Train/Test Split (70/30)**
   Partition data for model training and evaluation

Machine Learning Models Used

Logistic Regression

Linear classification with one-vs-rest strategy for multi-class problems

K-Nearest Neighbors

Instance-based learning using local neighborhood similarity

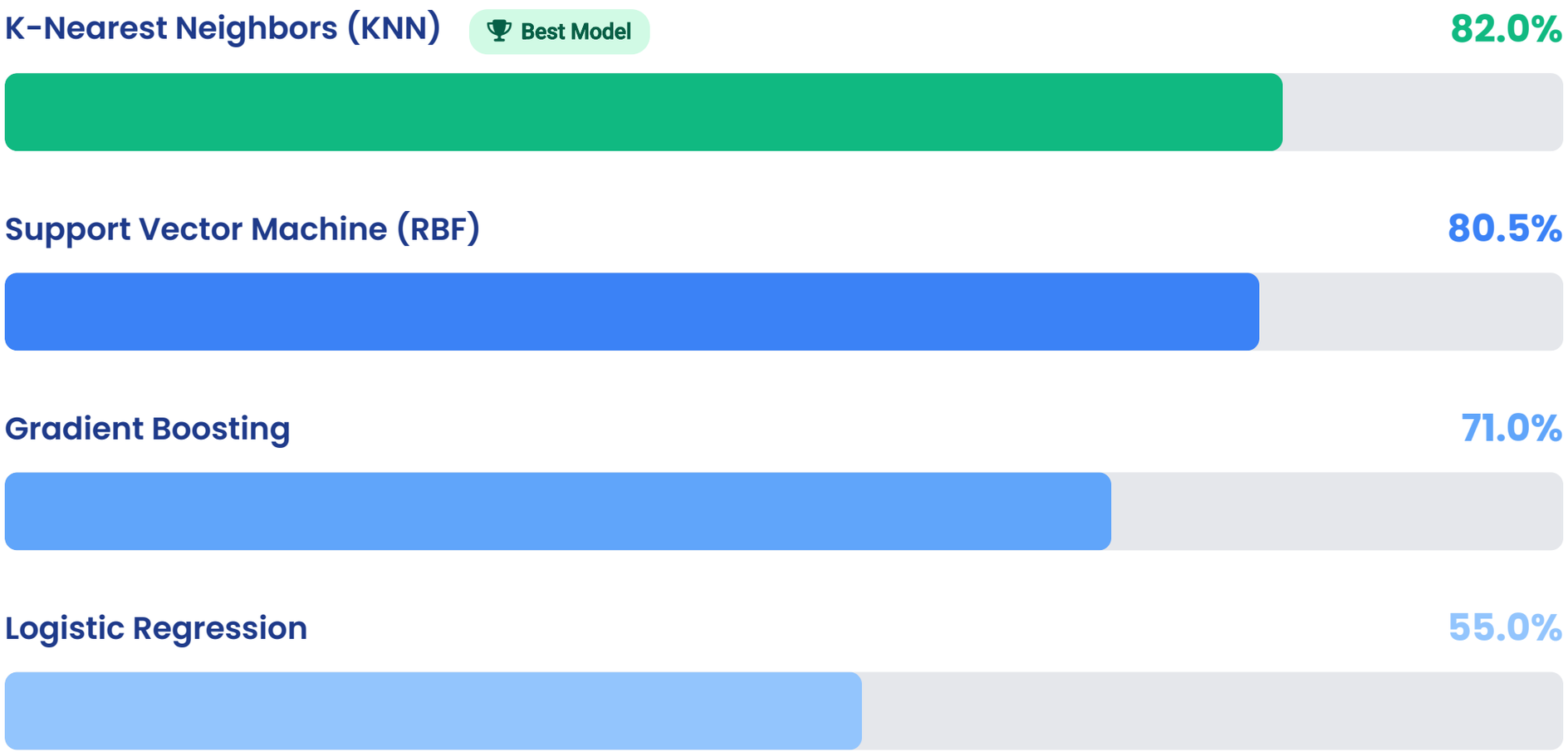
Support Vector Machine

Non-linear classification with RBF kernel for complex boundaries

Gradient Boosting

Ensemble method combining multiple weak decision tree learners

Results & Accuracy Comparison



KNN achieved the highest test accuracy at 82%, outperforming all other models

Performance Analysis

Top Performers

K-Nearest Neighbors (KNN) – 82%

Excels with clustered feature spaces by leveraging local neighborhoods, making it highly effective for sensor data classification

SVM (RBF Kernel) – 80.5%

Models complex non-linear decision boundaries effectively; achieves close second-place performance on this dataset

Key Insight

Both models handle the non-linear patterns in accelerometer and gyroscope data, with KNN benefiting from the natural clustering of activity classes

Lower Performers & Challenges

Logistic Regression – 55%

Limited by linear decision surface in complex, multi-dimensional sensor data; struggles with overlapping activity patterns

Gradient Boosting – 71%

Captures feature interactions but performs weaker on this particular feature set; decent ensemble performance overall

General Challenges

Sensor noise and class overlap between similar activities affect classification margins across all models

Conclusion

- Built a robust **HAR classifier** using wearable sensor data from smartphone and smartwatch devices
- Achieved **82% test accuracy** using K-Nearest Neighbors as the best performing model
- Multi-model comparison demonstrates **ML strengths and trade-offs** across different algorithms
- Applicable to **fitness tracking, health monitoring**, and context-aware applications

Future Work & Improvements

Deep Learning Models

Explore LSTM and CNN architectures to capture temporal patterns in sensor sequences

Live App Deployment

Deploy model to real-time smartwatch and smartphone applications

Expand Activity Classes

Extend beyond top 10 classes and include more diverse subjects and scenarios

Real-Time Visualization

Add live dashboards with on-device visualization and performance metrics